

SLIP – 4

Section I: Data Structures [15 Marks]

Q 1) Read the data from file 'student.txt' containing names of student and their class (FY, SY, TY). Accept a student name from user and use linear search algorithm to check whether the name is present in the file and output the Class of student otherwise output “Student not in the list”. Assumed file format (one record per line): Name Class e.g. Amit FY

Answer:

```
#include <stdio.h>
#include <string.h>

struct Student
{
    char name[50];
    char cls[5];
};

int main()
{
    FILE *fp;    struct
    Student s[100];    char
    key[50];    int n = 0, i,
    found = 0;

    fp = fopen("student.txt", "r");
    if(fp == NULL)
    {
        printf("File not found");
        return 0;
    }

    while(n < 100 && fscanf(fp, "%s %s", s[n].name, s[n].cls) == 2)
        n++;    fclose(fp);

    printf("Enter student name: ");
    scanf("%s", key);

    for(i = 0; i < n; i++)
    {
        if(strcmp(s[i].name, key) == 0)
        {
```

```

        printf("Class of %s is %s", key, s[i].cls);
found = 1;          break;
    }
}

    if(!found)      printf("Student not
in the list");

    return 0;
}

```

Section II: Database Management Systems [15 Marks]

Q.2) Consider the following database (primary keys are underlined).

Policy (pno, pname, premium_amt, policy_type)

Customer (cno, cname.city, agent_name)

The relationship is as follows: Policy-Customer: many-to-one. The ‘policy_type’ can have values as ‘Yearly’, ‘Half-yearly’ or ‘Monthly’.

Q 2(A)

A. Create the above database in PostGreSQL and insert sufficient records. [5 marks]

Answer:

```

CREATE TABLE Customer(
cno INT PRIMARY KEY,
cname VARCHAR(50),      city
VARCHAR(50),      agent_name
VARCHAR(50)
);

CREATE TABLE Policy(      pno INT
PRIMARY KEY,      pname
VARCHAR(50),      premium_amt
NUMERIC(10,2),      policy_type
VARCHAR(15)
CHECK(policy_type IN ('Yearly','Half-yearly','Monthly')),
cno INT REFERENCES Customer(cno)
);

INSERT INTO Customer VALUES
(1, 'Amit', 'Pune', 'Joshi'),
(2, 'Neha', 'Mumbai', 'Kale'), (3, 'Raj', 'Pune', 'Joshi');

```

```
INSERT INTO Policy VALUES
(11, 'Jeevan Anand', 5000, 'Yearly', 1),
(12, 'Jeevan Anand', 800, 'Monthly', 2),
(13, 'Jeevan Saral', 3000, 'Half-yearly', 1);
```

Q 2(B)

B. Execute the following queries in PostgreSQL (any 3) [6 marks]

i. List the details of all customers who live in city.

Replace <city_name> with the city specified by the examiner.

Answer:

```
SELECT * FROM Customer
WHERE city = '<city_name>';
```

ii. Display the average premium amount.

Answer:

```
SELECT AVG(premium_amt) FROM Policy;
```

iii. Count the number of customers who have taken 'Jeevan Anand' policy.

Answer:

```
SELECT COUNT(DISTINCT cno) FROM Policy
WHERE pname = 'Jeevan Anand';
```

iv. Increase the premium amount for 'Monthly' policies by 10%.

Q 2(C)

C. Write a query to list the policy_type with highest average premium amount. [4 marks]

Answer:

```
SELECT policy_type
FROM Policy
GROUP BY policy_type
HAVING AVG(premium_amt) >= ALL
(
    SELECT AVG(premium_amt)
    FROM Policy
)
```

```
GROUP BY policy_type  
);
```

Q 3) Viva [5 marks]